

Key assumptions (stated explicitly)

- **Interpretation of “top size”** — I treat the user’s “~7”” as the crusher *product* $P_{80} \approx 178 \text{ mm}$ and examine the effect of increasing the *product top size* (P_{80}) to **203 mm (8")** and **229 mm (9")**. If you meant feed top size (F_{80}) instead, tell me and I’ll re-run. (Design sheet shows ROM top size = 1200 mm; I used that as the feed-size for comminution math). The relevant values taken from your uploaded design file: ROM top size 1200 mm and crusher/product P_{80} 178 mm.
- **Sizing/work relationship** — I used **Bond-style crushing work relation** for relative crushing duty:

$$W \propto (1/P_{80} - 1/F_{80}) \propto \left(\frac{1}{\sqrt{P_{80}}} - \frac{1}{\sqrt{F_{80}}} \right)$$

This is the standard approach to estimate relative change in crushing work/power when P_{80} or F_{80} change. (See discussions of Bond crushing work & comminution fundamentals). ([911Metallurgist](#))

- **Cost → power scaling** — I assume **required installed/purchased crusher cost scales with installed power** and used a **cost-capacity exponent of 0.6** (the “six-tenths rule”):

$$\text{Cost}_2 = \text{Cost}_1 \times (\text{Power}_2 / \text{Power}_1)^{0.6} \quad \text{or} \quad \left(\frac{\text{Power}_2}{\text{Power}_1} \right)^{0.6}$$

This exponent is a standard cost estimating rule-of-thumb when detailed vendor quotes are not available. Use with $\pm 20\%$ caution. ([pdhonline.com](#), [Al-Mustaqbal University](#))

- **Base crusher purchase cost (2015)** — I used the 2015 gyratory crusher purchase cost from your uploaded cost tables (example: **60 in × 90 in gyratory** shown at **\$7,434,000 (2015)**).

- **Escalation to 2025** — your costs are 2015; I escalated nominal purchase costs to **2025** using an inflation factor consistent with CPI-based cumulative inflation from 2015→2025 ($\sim +36.3\%$ / multiplier **1.363**). This is a reasonable, conservative escalation for capital equipment; if you prefer ENR/CCI or regional escalation (steel, labour) I can swap indices. ([In 2013 Dollars](#), [Engineering News-Record](#))

- **Installed cost multiplier** — I used a **purchase→installed multiplier = 2.0** (i.e., installed cost $\approx 2\times$ purchased equipment cost). Adjust this multiplier to match your project specifics (site location, civil, foundations, structural work). If you have your project's detailed install factors in your Excel, I will use that. ($2\times$ is a common early-stage assumption for heavy rotating equipment including foundations, platforming, drives, electrical etc.). ([OSTI.gov](https://www.osti.gov/))

Calculations (steps shown, numbers rounded to sensible precision)

A. Bond-work relative change (using P80 = 1200 mm)

- Base P80 = 178 mm: $W_{base} \propto 1/178 - 1/1200 = 0.04609 W_{\{base\}}$ propto $1/\sqrt{178} - 1/\sqrt{1200} = 0.04609$ (arb. units).
 - P80 = 203 mm (8"): $W_8 \propto 0.04132 W_{\{8\}}$ propto 0.04132 → **W8/Wbase = 0.8966 ($\approx -10.3\%$ work)**.
 - P80 = 229 mm (9"): $W_9 \propto 0.03721 W_{\{9\}}$ propto 0.03721 → **W9/Wbase = 0.8075 ($\approx -19.25\%$ work)**.
- (So **increasing P80 reduces crushing work** — less reduction required). Source: Bond crushing relation / comminution references. ([911Metallurgist](https://www.osti.gov/))

B. Apply cost scaling (six-tenths rule)

- Base purchase cost (2015) for representative gyratory (60×90) = **\$7,434,000** (your cost table).
- Escalated to 2025: $\$7,434,000 \times 1.363 = \$10,132,542$ (purchase price 2025 \$). ([In 2013 Dollars](https://www.osti.gov/))
- Now scale by power-change^{0.6}:
 - **P80 = 8" (203 mm)** → purchase cost $\approx \$10,132,542 \times (0.8966)^{0.6} = \9.49M (purchase, 2025\$).
 - **P80 = 9" (229 mm)** → purchase cost $\approx \$10,132,542 \times (0.8075)^{0.6} = \8.91M (purchase, 2025\$).

C. Installed cost (using $\times 2$ multiplier)

- Installed, base (P80=178 mm): **$\$10.13\text{M} \times 2 = \20.27M** (2025\$).

- Installed, P80=8": $\$9.49\text{M} \times 2 = \18.98M .
- Installed, P80=9": $\$8.91\text{M} \times 2 = \17.83M .

D. Net TIC impact (examples)

Take three example project TICs (installed total): **\$50M, \$100M, \$200M** and compute percent change in TIC from the crusher change:

- **If TIC = \$50M:**
 - P80→8" reduces crusher-installed cost by ~\$1.05M → TIC **down** ~2.6%.
 - P80→9" reduces crusher-installed cost by ~\$2.34M → TIC **down** ~4.9%.
- **If TIC = \$100M:** reductions $\approx$ -1.3% (8") and -2.34% (9").
- **If TIC = \$200M:** reductions $\approx$ -0.64% (8") and -1.22% (9%).

(Explanation: absolute installed-crusher savings are ~\$1.05M (to 8") and ~\$2.34M (to 9") in 2025\$; percent TIC impact = saving ÷ project TIC). The math above used your 2015 cost tables escalated to 2025 and the six-tenths rule.

Interpretation / Engineering context

- **Why the reduction occurs:** increasing the allowed product top size (= making the crushed product coarser) reduces the crushing reduction ratio and therefore reduces the mechanical work required; that lowers required crusher power and allows a lower-capacity/power machine or lower motor HP specification. That cascades to lower purchase price and smaller civil/structural requirements. (See Bond crushing relation and Metso/handbook notes on crusher sizing). ([911Metallurgist](#), [metso.com](#))
- **But: real projects are more complex.** The crusher is only one element of the comminution train; a coarser primary product may **shift burden to downstream equipment (secondary/tertiary crushing or SAG/Ball mill)** or change grinding performance, recovery, and downstream processing costs. If by increasing P80 you avoid a secondary crusher or reduce mill load, the net TIC and OPEX benefits could be larger than computed above; conversely, if the coarser primary product forces additional screening/handling or changes throughput constraints, savings may be less. You must evaluate the whole comminution circuit & economics, not only the primary crusher item. (Comminution normally dominates processing OPEX; crushing typically shares a non-trivial percent of capex/opex). ([MDPI](#), [metso.com](#))
- **OPEX effects:** reduced crushing work reduces electrical energy consumption and long-term operating cost. For example, lower crushing power reduces

energy use (kW), lowering OPEX — over years this can be material. Estimate OPEX savings by converting % reduction in crushing work to kWh/year and applying local energy cost.

Sources & files used (key load-bearing citations)

- Your uploaded **Gyratory Crusher Design Criteria** (design parameters: ROM top size 1200 mm, crusher OSS/P80 178 mm etc.).
- Your uploaded **2015 Gyratory Crusher Cost Tables (InfoMine / Milling & Processing Equipment)** — used the 60"×90" gyratory cost \$7,434,000 (2015).
- Bond crushing / comminution references (Bond crushing work concepts). ([911Metallurgist](#))
- Six-tenths rule / cost scaling references. ([pdhonline.com](#), [Al-Mustaqbal University](#))
- CPI/Inflation conversion 2015→2025 (used ~+36.3% cumulative). ([In 2013 Dollars](#), [US Inflation Calculator](#))
- Metso / crushing handbook references on crusher selection and product sizing. ([metso.com](#))

Practical recommendations / next steps (what I would do as a design engineer / estimator)

- **Confirm exactly which “top size” you mean** (feed top size F80 or product P80). I assumed product P80; if you intended feed, I will re-run with that input.
- **Use your project’s actual crusher size / purchase + install factors from the Excel capex** (you uploaded an Excel — I can ingest it and re-run using the exact crusher purchase & installation multipliers you used). That will produce more accurate TIC % impacts than my generic ×2 multiplier. (I can ingest your .xlsx now if you want.)
- **Run whole-circuit sensitivity** — evaluate how the changed P80 affects secondary crushing requirements, mill throughput & energy, and recovery (NPV impact). Sometimes increasing primary P80 reduces capital today but increases downstream costs; run a case-level NPV comparison.
- **Ask vendors for two quotes** for crushers sized/quoted for each P80 (this will capture real vendor selection differences, mechanical ancillaries and shipping costs). Vendor quotes often differ substantially from scaled estimates.

- **Capture OPEX benefit** — convert the % reduction in crushing work to kW and \$/year to estimate NPV of operational savings. That will often improve the economics beyond the simple TIC delta.
- **Document the uncertainty** — the six-tenths rule and Bond-work approximations are reliable for early-stage scoping ($\pm 20\%$); flag this in reports.

Quick numeric summary (rounded, 2025\$)

- **Base purchase (2015 table) $\Rightarrow$ 2025 escalated purchase: \$7.43M $\rightarrow$ \$10.13M.**
- **Purchase at P80 = 8" (203 mm): \$9.49M (2025\$).**
- **Purchase at P80 = 9" (229 mm): \$8.91M (2025\$).**
- **Installed ($\times 2$ multiplier): Base \$20.27M $\rightarrow$ 8" \$18.98M $\rightarrow$ 9" \$17.83M.**
- **TIC impact (example TIC = \$50M): P80 $\rightarrow$ 8" reduces TIC $\approx -2.6\%$; P80 $\rightarrow$ 9" reduces TIC $\approx -4.9\%$. For a \$200M project the impacts are $\sim -0.64\%$ and -1.22% respectively. (Absolute installed-crusher savings: $\sim \$1.05\text{M} \rightarrow \2.34M).**

Final note on uncertainty

These numbers are an **early-stage engineering estimate** based on: (a) 2015 vendor purchase cost table you supplied, (b) Bond work and 0.6 cost exponent, and (c) CPI escalation. Expect $\pm 20\text{--}30\%$ on the crusher cost estimates; the **direction** (that raising P80 reduces crusher size, power and cost) is robust. The **net project TIC** impact depends strongly on what fraction of processing capex your primary crusher occupies and the downstream effects on grinding and recovery — that's why I recommend a short whole-circuit NPV sensitivity for the three P80 cases.

If you'd like I can immediately:

- **(A) Open your uploaded Excel capex** and re-run these calculations using the exact crusher purchase/installation factors and your project TIC line items, and produce a downloadable spreadsheet with the delta analysis; **or**
- **(B) Produce a quick two-page PDF** summarizing the inputs, calculations, and sensitivity tornado chart (P80 vs TIC impact), suitable for your estimator / investor reporting.

Which would you prefer?

This is where Chatgpt was wrong in that it did not change the size of the crusher and iterated on the same crusher and extrapolated the costs. In reality, with changing the OSS, a different crusher will have to be selected all together that will drastically change the capex. Also chatgpt did not read the lang factors in the spreadsheet and assumed incorrect factor (2X) for installation cost.

1) What I changed vs the earlier pass

- **I selected alternate machine families** rather than holding the same gyratory frame constant.
- **I used your guidance on installation factors:** applied **3.5× equipment purchase cost** as a representative installed/TIC multiplier for the crusher + material-handling scope (foundations, steel, conveyors, drives, electrical, piping, instrumentation, indirects & owner's cost). You had suggested 3–4×; I used 3.5× as a mid-point.
- **Kept the same comminution work relationships** (Bond-style P80 relation) to estimate relative power change when P80 goes from 178 mm → 203 mm → 229 mm.
- **Sourced vendor & cost references** (Metso crushing handbook for crusher selection rules; jaw/gyratory price ranges; cost-estimating descriptions and factor estimating sources). ([metso.com](https://www.metso.com), [zenithcrusher.com](https://www.zenithcrusher.com), [costengineering.eu](https://www.costengineering.eu))

2) Key engineering suitability constraints (don't skip these)

- **Throughput / capacity** is the main gating factor. Gyratory crushers are the standard for *very high throughput* primary crushing (large ROM size and continuous high tph). Primary jaw crushers can be economical at smaller throughputs but become impractical for very large operations due to capacity, wear, and uneven feed handling. Vendor catalogues and handbooks must be used to check tph vs gape vs model. ([metso.com](https://www.metso.com), [AusIMM](https://www.ausimm.com))
- **ROM top size and feed handling:** If ROM top size is very large (you

earlier noted ROM 1200 mm), a primary jaw or apron feeder + grizzly may be necessary ahead of certain crushers — again impacting install scope and costs.

- **Downstream impact:** A coarser P80 can shift duty to downstream crushers/grinding; net project TIC/NPV must be considered for the whole circuit, not only primary crusher cost.

3) Assumptions used for the numbers below (explicit)

- **Baseline purchase cost (2015 table):** used the same gyratory purchase from your file, escalated to 2025 earlier (purchase $\approx$ US\$10.13M nominal in 2025\$ for the representative gyratory).
- **Escalation / indices:** used CPI-style escalation to 2025 (same as earlier). (metso.com)
- **Relative crushing work (Bond-style):** increasing product P80 to 203 mm (8") reduces crushing work by $\sim 10.3\%$; to 229 mm (9") reduces it by $\sim 19.3\%$ vs baseline 178 mm. (Same comminution relation used previously). (metso.com, Carnegie Mellon University)
- **Cost scaling with power:** applied the six-tenths rule ($\text{cost} \propto \text{power}^{0.6}$) to reflect that a lower power requirement usually maps to a lower equipment purchase cost. (costengineering.eu)
- **Installed/TIC multiplier: $3.5\times$ purchase cost** used to estimate crusher + material handling installed line items (foundations, conveyors, supports, electrical, instrumentation, erection, commissioning, indirects, and owner's soft costs). You instructed $3\text{--}4\times$, so I used $3.5\times$ midpoint.
- **Alternative purchase cost ranges** (because vendor lists vary): I used two plausible alternatives consistent with vendor catalogs and marketplace pricing:
 - *Smaller gyratory frame* (reasonably $\sim 10\text{--}20\%$ lower purchase cost than baseline after power reduction) — modeled as **purchase $\approx 0.85\text{--}0.90 \times$ baseline** after scaling.
 - *Large primary jaw alternative* (a robust, mining-grade jaw for substantial throughput): modeled as **purchase $\approx \text{US}\$1.5\text{--}3.5\text{M}$** (purchase price range for very large jaw units from supplier price lists & market listings — jaw prices vary greatly with vendor, capacity and are typically much lower than large gyratory frames). See vendor price guidance. (zenithcrusher.com, Machinio)

Important: these jaw price numbers are indicative for very large jaw models; for truly very high capacities (multi-thousand tph), giant gyratory machines are often required, and a jaw option may not be technically viable.

4) Calculations — headline numbers (2025\$), using 3.5× installed factor

A. Baseline (original gyratory, escalated)

- **Baseline purchase (esc to 2025): US\$10.13M** (from your table).
- **Installed (3.5×): US\$35.46M.**

B. Option 1 — *Smaller gyratory frame* (realistic if product P80 relaxed)

- **Assumed purchase:** $0.85\text{--}0.90 \times \text{baseline purchase} \rightarrow \text{US\$8.61M} - \text{US\$9.12M}.$
- **Installed (3.5×): US\$30.14M – US\$31.92M.**
- **Installed saving vs baseline: US\$4.5M – US\$4.9M ($\approx 12.6\% - 13.8\%$ reduction in installed crusher line).**

C. Option 2 — *Large primary jaw alternative* (if technically viable)

- **Assumed purchase:** US\$1.5M – US\$3.5M (range for very large jaw units; depends on vendor & spec). (zenithcrusher.com, Machinio)
- **Installed (3.5×): US\$5.25M – US\$12.25M.**
- **Installed saving vs baseline: US\$23.2M – US\$30.2M ($\approx 65\% - 85\%$ reduction in installed crusher line) — *if jaw is acceptable for capacity/operation.***

5) Translate to project TIC impact (examples)

Take three illustrative project TICs to show percent effects (you can substitute your project TIC):

- **Project TIC = US\$50M**
 - Baseline crusher installed = US\$35.46M $\rightarrow$ would represent **71%** of the project TIC (this shows the caution: using 3.5× applied solely to the crusher purchase may produce very large installed numbers if the reference purchase is of a very large machine — in practice, crusher + material handling scope is often a

significant chunk but rarely >50% of entire plant).

- Smaller gyratory → installed ~US\$31M → TIC reduction ~US\$4.5M → ~9% TIC decrease.
- Jaw alternative → installed ~US\$5–12M → TIC reduction ~US\$23M–30M → ~46–60% TIC decrease (but jaw must be feasible).
- **Project TIC = US\$100M**
 - Smaller gyratory → TIC down ~4.5%.
 - Jaw alternative → TIC down ~23–30%.
- **Project TIC = US\$200M**
 - Smaller gyratory → TIC down ~2.25%.
 - Jaw alternative → TIC down ~11.5–15%.

Interpretation: the **absolute dollar** saving from selecting a different crusher can be large; the **percentage of TIC** saved depends strongly on the project size. Your earlier note is exactly right: using a 3.5× installed multiplier magnifies the equipment purchase differences into large TIC deltas — but you must check that the installed multiplier is being applied consistently (e.g., are conveyors, stockpile, reclaim, and major civil scopes included in the same line or in separate line items in the capex spreadsheet?). If the purchase base is a very large gyratory, the installed line for the crushing complex can dominate a smaller project — which is plausible for greenfield mills.

6) Engineering caveats & final checks (must do these before acting on these deltas)

- **Verify throughput & vendor capacity curves:** determine the tph target. If tph > vendor jaw capacity at the required feed size, jaw is *not* a realistic alternative. Use Metso / Sandvik / FLS catalogs to confirm model capacities vs gape/discharge. (metso.com, AusIMM)
- **Check feed handling & grizzly/upstream works:** some jaw options require different feeders, grizzlies, or primary screening which add to installed cost.
- **Downstream comminution balance:** coarser P80 may increase mill energy or require extra crushing stages; run a whole-circuit energy + costing check to get net NPV effect.
- **Itemization in your Excel capex:** your second sheet reportedly has lang/installation factors — use those per-category factors to compute installed costs per line item instead of a blanket 3.5×. That will give realistic installed numbers (e.g., foundations & structural may have different factors than conveyors or electrical).
- **Ask vendor quotes:** early vendor indicative quotes will almost always be the tie-breaker and can differ materially from factor estimates.

7) Sources and where I pulled references from

- **User-supplied:** Gyratory Crusher Cost Tables (2015) and Gyratory Crusher Design Criteria.
- **Vendor handbooks & catalogs:** Metso Crushing & Screening handbook (capacity, crusher selection guidance). (metso.com)
- **Jaw crusher price guidance & market listings:** jaw price guides and used listings (illustrative price ranges). (zenithcrusher.com, Machinio)
- **Cost estimating / factor estimating sources:** equipment factored estimating guidance and installation factor discussion. (costengineering.eu)

8) Recommended next steps (practical, prioritized)

- **I will re-open your Excel (the “Capital Estimate Worksheet” and the “Estimated Connected Power” tabs) and use *your* installation/lang factors per category** to re-compute installed/TIC values for each crusher candidate — that will remove the blunt $3.5\times$ assumption and apply the exact multipliers you already prepared. (I attempted to read the Excel earlier but had a tool timeout; I can retry immediately.)
- **Collect 2–3 vendor sizing & indicative price points** for each candidate crusher (original frame, smaller gyratory, and a very large jaw option) from Metso / Sandvik / thyssenkrupp / FLS and map purchase + delivery lead times. I’ll cite those vendor pages.
- **Run a whole-circuit sensitivity:** show net TIC + 20-yr OPEX delta for the three P80 cases (178, 203, 229 mm) and the three equipment choices (baseline gyratory, smaller gyratory, jaw alternative). This will show whether the primary crusher selection yields net project savings after downstream changes.
- Deliver a spreadsheet and short report with exact line-item deltas so you can present to stakeholders or to vendors for official quotes.

Quick answer to your original ask (concise)

- **Yes — selecting a different crusher type can materially change the TIC.** Using your guidance ($3\text{--}4\times$ installed factor), a realistic smaller gyratory saves **~US\$4–5M installed** vs the baseline (for the device studied), while a jaw alternative *could* save **US\$20–30M installed** — *but only if the jaw is technically acceptable for the*

required throughput and ROM. If the jaw isn't suitable, the smaller gyratory is a safer option producing modest but meaningful TIC savings. Exact numbers must be recomputed using your Excel's category-specific multipliers and vendor curves for accuracy.

If you want, I'll **immediately**:

- (A) re-open your Excel and re-run the installed cost calculations using the *exact* language/installation factors on the second tab, and
- (B) fetch vendor sizing/price reference points and produce a downloadable spreadsheet + short PDF summary with the per-line-item deltas and whole-circuit sensitivity.

Which should I do first — re-run the Excel with your project's factors, or gather vendor model/price references? (I can start both in parallel if you prefer.)